Thermo chemistry

What is thermochemistry?

The branch of physical chemistry which deals with the thermal or heat changes caused by chemical reactions is called thermochemistry

Every substance has a definite amount of energy that is intrinsic energy or internal energy, E.

Unit: $1\ J = 1\ kg \cdot m^2 / s^2$

$1\ cal = 4.184\ J$

What is Enthalpy of reaction (H)/ Heat of reaction

The enthalpy of reaction is ordinarily defined as the amount of heat evolved or absorbed when the reaction has taken place at constant pressure between the numbers of moles of the reactants as shown in the balanced equation.

$C\ (s) + O_2\ (g) \longrightarrow CO_2\ (g)$; $\Delta H = -393.5\ KJ/mol$

Exothermic reactions.

1. An exothermic reaction is a reaction in which heat is released by the system to the surroundings. From the perspective of the system, "heat is given off."
2. For an exothermic reaction, the sign of ΔH is negative.
3. When a reaction is exothermic (ΔH is negative), that is a favorable condition.

ΔH = heat of product – heat of reactant = 100 kj – 110 kj = -10 kj

Endothermic reactions

1. An endothermic reaction is a reaction in which heat is absorbed by the system from the surroundings. From the perspective of the system, "heat is taken in."
2. For an endothermic reaction, the sign of ΔH is positive.

3. When a reaction is endothermic (ΔH is positive), that is an unfavorable condition.

ΔH = heat of product – heat of reactant = 100 kj – 90 kj = +10 kj

The presentation of exothermic and endothermic enthalpy diagram

Thermochemical equation with an example

An equation which indicates the amount of heat change (evolved or absorbed) in the reaction or process is called a Thermochemical equation.

It must essentially : (*a*) be balanced; (*b*) give the value of ΔE or ΔH corresponding to the quantities of substances given by the equation; (*c*) mention the physical states of the reactants and products. The physical states are represented by the symbols (*s*), (1), (*g*) and (aq) for solid, liquid, gas and gaseous states respectively.

Example of Thermochemical Equation

$$H_2(g) + \tfrac{1}{2} O_2(g) \longrightarrow H_2O(l) \qquad \Delta H = -68.32 \text{ kcal}$$

HEAT OF FORMATION

The heat of formation of a compound is defined as :

The change in enthalpy that takes place when one mole of the compound is formed from its elements.

It is denoted by ΔH_f. For example, the heat of formation of ferrous sulphide and acetylene may be expressed as :

$$Fe(s) + S(s) \longrightarrow FeS(s) \qquad \Delta H_f = -24.0 \text{ kcal}$$

$$2C(s) + H_2(g) \longrightarrow C_2H_2(g) \qquad \Delta H_f = +53.14 \text{ kcal}$$

HEAT OF COMBUSTION

The heat of combustion of a substance is defined as : **the change in enthalpy of a system when one mole of the substance is completely burnt in excess of air or oxygen.**

It is denoted by ΔH_c. As for example, heat of combustion of methane is – 21.0 kcal (= 87.78 kJ) as shown by the equation

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l) \qquad \Delta H_c = -21.0 \text{ kcal}$$

HEAT OF SOLUTION

the change in enthalpy when one mole of a substance is dissolved in a specified quantity of solvent at a given temperature.

$$KCl(s) + H_2O(l) \longrightarrow KCl(aq) \qquad \Delta H = -4.4 \text{ kcal}$$

Hess's law of constant heat summation

Statement: If a chemical change can be made to take place in two or more different ways whether in one step or two or more steps, the amount of total heat change is same no matter by which method the change is brought about.

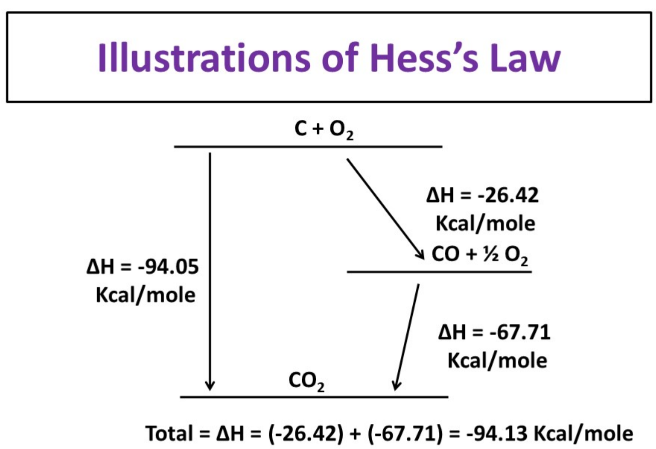

Fig: Illustration of Hess's law

Applications of Hess's law

1. Determination of heat of formation of substances which otherwise cannot be measured experimentally.
2. Determination of heat of transition
3. Determination of heat of various reactions

Problem: Calculate the heat of formation of acetic acid from the following information:

(1) $CH_3COOH\ (l) + 2O_2\ (g) \longrightarrow 2\ CO_2\ (g) + 2H_2O\ (l)$; $\Delta H^o = -871.5\ kJmol^{-1}$

(2) $C\ (s) + O_2\ (g) \longrightarrow CO_2\ (g)$; $\Delta H^o = -393.5\ kJmol^{-1}$

(3) $H_2\ (g) + 1/2\ O_2\ (g) \longrightarrow H_2O\ (l)$; $\Delta H^o = -285.85\ kJmol^{-1}$

Solution:

Multiply equation 2 and 3 by 2 and add them

$2\ C\ (s) + 2O_2\ (g) \longrightarrow 2CO_2\ (g)$; $\Delta H^o = -787.0\ kJmol^{-1}$

$2\ H_2\ (g) + O_2\ (g) \longrightarrow 2H_2O\ (l)$; $\Delta H^o = -541.7\ kJmol^{-1}$

(4) $2\ C\ (s) + 2\ H_2\ (g) + 3O_2\ (g) \longrightarrow 2CO_2\ (g) + 2H_2O\ (l)$; $\Delta H^o = -1358.7\ kJmol^{-1}$

If we subtract equation 1 from equation 4, then we will get

$2\ C\ (s) + 2\ H_2\ (g) + 3O_2\ (g) \longrightarrow 2\cancel{CO_2}\ (g) + 2\cancel{H_2O}\ (l)$; $\Delta H^o = -1358.7\ kJmol^{-1}$

$CH_3COOH\ (l) + 2O_2\ (g) \longrightarrow 2\ \cancel{CO_2}\ (g) + 2\cancel{H_2O}\ (l)$; $\Delta H^o = -871.5\ kJmol^{-1}$

$2\ C\ (s) + 2\ H_2\ (g) + O_2\ (g) \longrightarrow CH_3COOH\ (l)$; $\Delta H^o = -487.2\ kJmol^{-1}$

So, according to Hess's law of heat summation, the heat of formation of acetic acid is -487.2 $kJmol^{-1}$.

Write Lavoisier and Laplace's law with an example

The heat change accompanying a chemical reaction in one direction is exactly equal in magnitude, but opposite in sign, to that associated with the same reaction in the reverse direction. This law states that; the heat change (or enthalpy change) of a chemical reaction is exactly equal but opposite in sign for the reverse reaction. This is evident from the following two fractions:

(a) $CH_4\ (g) + 2O_2\ (g) \rightarrow CO_2\ (g) + 2H_2O\ (l)$ [here, $\Delta H^0 = -890.3\ kJ\ mol^{-1}$]

(b) $CO_2\ (g) + 2H_2O\ (l) \rightarrow CH_4\ (g) + 2O_2\ (g)$ [here, $\Delta H^0 = +890.3\ kJ\ mol^{-1}$]

Thus, it can be concluded that, $\Delta H_{\text{forward reaction}} = -\ \Delta H_{\text{backword reaction}}$

SOLVED PROBLEM 2. The heat of combustion of ethyl alcohol is – 330 kcal. If the heat of formation of $CO_2(g)$ and $H_2O(l)$ be – 94.3 kcal and – 68.5 kcal respectively, calculate the heat of formation of ethyl alcohol.

SOLUTION

We are given

(*a*) $C_2H_5OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l) \quad \Delta H = -330.0 \text{ kcal}$

(*b*) $C(s) + O_2(g) \longrightarrow CO_2(g) \quad \Delta H = -94.3 \text{ kcal}$

(*c*) $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l) \quad \Delta H = -68.5 \text{ kcal}$

We have to manipulate these equations so as to get the required equation :

$$2C(s) + 3H_2(g) + \tfrac{1}{2}O_2(g) \longrightarrow C_2H_5OH(l) \qquad \Delta H = ?$$

where ΔH is the heat of formation of ethyl alcohol.

Multiplying equation (*b*) by 2 and equation (*c*) by 3 and adding up these, we get

$$2C(s) + 3H_2(g) + \tfrac{3}{2}O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$$
$$\Delta H = -394.1 \text{ kcal}$$

Subtracting equation (*c*) from the above equation, we have

$$2C(s) + 3H_2(g) + \tfrac{1}{2}O_2(g) \longrightarrow C_2H_5OH(l)$$
$$\Delta H = -394.1 - (-330.0)$$

Thus the heat of formation of ethyl alcohol is **– 64.1 kcal.**

SOLVED PROBLEM. Calculate the standard heat of formation of propane (C_3H_8) if its heat of combustion is – 2220.2 kJ mol^{-1}. The heats of formation of $CO_2(g)$ and $H_2O(l)$ are – 393.5 and – 285.8 kJ mol^{-1} respectively.

SOLUTION

We are given

(*i*) $C_3H_8(g) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(l) \quad \Delta H_c = -2220.2 \text{ kJ}$

(*ii*) $C(s) + O_2(g) \longrightarrow CO_2(g) \quad \Delta H = -393.5 \text{ kJ}$

(*iii*) $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l) \quad \Delta H = -285.8 \text{ kJ}$

We should manipulate these equations in a way so as to get the required equation

$$3C(s) + 4H_2(g) \longrightarrow C_3H_8(g) \qquad \Delta H = ?$$

Multiplying equation (*ii*) by 3 and equation (*iii*) by 4 and adding up we get

$$3C(s) + 3O_2(g) \longrightarrow 3CO_2(g) \qquad \Delta H = -1180.5 \text{ kJ}$$
$$4H_2(g) + 2O_2(g) \longrightarrow 4H_2O(l) \qquad \Delta H = -1143.2 \text{ kJ}$$

(*iv*) $3C(s) + 4H_2(g) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(l) \quad \Delta H = -2323.7 \text{ kJ}$

Subtracting equation (*i*) from equation (*iv*), we have

$$3C(s) + 4H_2(g) + 5O_2(g) - 5O_2(g) \longrightarrow C_3H_8(g) \qquad \Delta H = -103.5 \text{ kJ}$$

$\therefore$ The heat of formation of propane is **– 103.5 kJ mol^{-1}.**